MEREDITH STONE

meredithstone@arizona.edu | astronomeredith.github.io | 0000-0002-9720-3255

EDUCATION

University of Arizona 2022 - present
Ph.D. candidate; defending April 2027
NSF Graduate Research Fellow
Advisor: Prof. George Rieke, Dr. Stacey Alberts (STScI)

University of Massachusetts Amherst 2018 - 2022
B.S. Astronomy and Physics, *summa cum laude*

RESEARCH OVERVIEW

I have extensive scientific expertise in JWST data reduction and analysis, including high-precision techniques such as NIRCam point-spread function subtraction and MIRI LRS slit-stepping. My interests span from the hosts and environments of high-redshift quasars, to dust and the interstellar medium at cosmic noon, to PAH science with NASA's upcoming PRIMA mission.

Publications: 20 refereed publications, 5 first-author; 4 in prep/submitted publications, 1 first-author.
1300+ total citations, 175+ on first-author publications, h-index: 12; full list below.

Conferences and Talks: 9 science talks including 2 invited talks; full list below

RESEARCH APPOINTMENTS

University of Arizona, Steward Observatory Tucson, AZ
Graduate Researcher and NSF GRFP Fellow *May 2022 - present*

Yale University, Department of Astronomy New Haven, CT
Dorrit Hoffleit Undergraduate Intern *May - July 2021*

University of Massachusetts Amherst, Department of Astronomy Amherst, MA
Student Researcher *January 2019 - May 2022*
FCAD Summer Research Program Intern *May - July 2019*

FELLOWSHIP AND GRANT FUNDING

National Science Foundation Graduate Research Fellowship 2024 - 2027
Prize Fellowship, totaling $\sim$ \$229,000 for three years of graduate stipend and tuition

Dorrit Hoffleit Undergraduate Research Scholarship 2021
Yale University, totaling $\sim$ \$5000 for a summer of undergraduate research

Commonwealth Honors College Research Assistant Fellowship 2020 - 2021
University of Massachusetts Amherst, totaling $\sim$ \$2000 for in-semester research funding

Massachusetts Space Grant Fellowship & David J. Van Blerkom Research Scholarship 2019
University of Massachusetts Amherst, totaling ∼ $4000 for a summer of undergraduate research

HONORS AND AWARDS

2026: University of Arizona College of Science Galileo Circle Scholarship
2025: Steward Observatory/Department of Astronomy Graduate Student Award in Service
2022: NSF Graduate Research Fellowship Honorable Mention
2022: Five College Astronomy Department Mary Dailey Irvine Thesis Prize
2022: UMass Amherst Department of Astronomy Outstanding Senior Award
2022: UMass Amherst Physics Philip Hasbrouck Scholarship for Academic Excellence
2021: Goldwater Scholarship
2021: AAS Chambliss Astronomy Achievement Student Award - Honorable Mention

TEACHING AND MENTORING EXPERIENCE

University of Arizona

- Graduate Coordinator for TIMESTEP Research Apprenticeship (Fall 2023 - present)
  - One of two initial lead graduate coordinators for the Research Apprenticeship, a research skills development and job training program for physics/astronomy sophomores
  - Created and delivered weekly workshops on hard and soft research skills not typically taught in physics/astronomy curriculum
  - Served as outside-of-research-group mentor to 4-5 students per cohort, 18 total over 4 years
  - Grew the program from a cohort of 10 students and 2 graduate coordinators in 2023-2024 to 24 students and 5 graduate coordinators in 2026-2027.
  - Created University of Arizona Space Sciences Research Map to help undergraduates identify potential research mentors in overwhelmingly large University of Arizona space sciences enterprise
- Teaching Assistant for ASTR 276 Research Methods in Astronomy pilot program (Fall 2026)
  - Helped adapt TIMESTEP Research Apprenticeship materials into new ASTR 276 class open to all students, taught by Profs. Gurtina Besla and Ed Prather
  - Taught 10 lectures in class's first semester offered, covering Python, LaTeX, research notes, Unix, DS9, and preparing a resume/CV for future research applications
- Graduate Coordinator for TIMESTEP Summer Tech Internship (Summer 2024)

- Teaching Assistant for ASTR 418/518 Astronomical Instrumentation (Fall 2024)

University of Massachusetts Amherst

- Undergraduate teaching assistant for PHYS 131/132: mechanics, optics, waves for non-physics majors (Fall 2019 - Spring 2022, including Summer 2020)
  - Worked directly with students in team-based learning format and during office hours in Physics Help Room
  - In Summer 2020, worked as sole teaching assistant for fully asynchronous summer session class of $\sim$ 100 students; served as students' primary contact running office hours, help sessions, and exam review sessions.

OUTREACH, COMMUNITY ENGAGEMENT, AND SERVICE

Academic and Professional Service

- Local Organizing Committee member for 2027 RiekeFest conference at University of Arizona (ongoing)
- Steward Observatory Graduate Student Council (ongoing) - I serve as a liason between the graduate students and department leadership regarding stipends, the academic program, and department culture
- Referee for *The Astrophysical Journal* (ongoing)
- Steward Observatory prospective visit planning committee leader (2025)
- Graduate student representative for 2025 Astronomy/Steward Observatory Academic Program Review

Public Outreach

- Arizona Astronomy Camp counselor (Summer 2024, 2025, 2026) - I mentor research projects during one-week camps for teenagers, prepare and lead activities on galaxy classification and evolution, and operate 32" and 61" telescopes for camper observations
- Southern AZ Regional Science and Engineering Fair judge (ongoing)
- Space Drafts (Tucson Astronomy on Tap) public talk (January 2025)

Research in the Media

- *Spotting the Difference: Supermassive Black Holes in Cosmological Simulations Don't Match Observations* for astrobites.org (2023)
- Expert Interview, U.S. News Brandfuse: Wonder Makes Us Investigate the First Light of the Universe (2022)
- *Understanding the Balance of Star Formation and Black Holes in Nearby Luminous Infrared Galaxies* for astrobites.org (2021)

SCIENTIFIC PRESENTATIONS

Invited Talks (2):

2 Steward Observatory EURECA Seminar, December 2025, Tucson AZ

1 MIRI team symposium, March 2024, Tucson AZ

Contributed Talks (7):

7. Charting Cosmic Dawn in Copenhagen, April 2026, Helsingor, Denmark
6. AAS 247th Winter Meeting, January 2026, Phoenix, AZ
5. NIRCam instrument team meeting, October 2025, Tucson, AZ
4. Building Physical Simulations of PAH Emission in Galaxies, November 2024, UVA, Charlottesville, VA
3. EREBUS collaboration meeting, October 2023, Tucson, AZ
2. 2022 New England Regional Quasar and AGN Meeting, May 2022, UConn, Storrs, CT
1. 2021 Conference for Undergraduate Women in Physics, January 2020, Yale University, New Haven, CT

Posters (6):

6. Exploring the Aromatic Universe in the JWST Era, July 2026, London, Canada
5. Pan-Dust, November 2025, Tucson, AZ
4. Infrared Spectroscopy from Space, October 2025, IPAC, Pasadena, CA
3. First Year of JWST Science, October 2023, STScI, Baltimore, MD
2. AAS 240th Summer Meeting, June 2022, Pasadena, CA
1. AAS 237th Winter Meeting, January 2021, virtual

PUBLICATION LIST

20 refereed publications, 5 first-author; 4 in prep/submitted publications, 1 first-author. 1300+ total citations, 175+ on first-author publications. h-index: 12.

In preparation or under review:

1. **Stone, M.** et al., *A MIRI LRS Study of the PAH-CO Relation in Main Sequence Galaxies at* $z \sim 1$. In prep, to be submitted to ApJ by conclusion of PhD.
2. Sun, Y., **Stone, M.** et al., *The Evolution of the* $M_{\rm BH}/M_*$ *Relation Revisited.* In prep, to be submitted to ApJ.

3. Vanicek, V., ..., **Stone, M.** et al. 2026, *Star Formation and Nebular Attenuation from Paα, Brα and Brβ in Massive Dust-obscured Galaxies at Cosmic Noon using JWST.* arXiv:2609.22428, under review in ApJ.

4. Zhu, Y., ..., **Stone, M.** et al. 2026, *PhotoIFU: NIRCam as a Photometric Integral Field Unit for Mapping Feedback in Galaxies.* arXiv2607.23020, under review in ApJ.

First author:

20. **Stone, M.**, Rieke, G., Lyu, J., et al., 2025, *The $z = 7.08$ Quasar ULAS J1120+0641 May Never Reach a "Normal" Black Hole to Stellar Mass Ratio.* ApJ 993.168

19. **Stone, M.,** Alberts, S., Rieke, G., et al., 2024, *5-25 μm Galaxy Number Counts from Deep JWST Data.* ApJ 972.62

18. **Stone, M.,** Lyu, J., Rieke, G., Alberts, S., and Hainline, K., 2024, *Undermassive Host Galaxies of Five $z \sim 6$ Luminous Quasars Detected with JWST.* ApJ 964.90

17. **Stone, M.,** Lyu, J., Rieke, G., and Alberts, S., 2023, *Detection of the Low-Stellar Mass Host Galaxy of a $z \sim 6.25$ Quasar with JWST.* ApJ 953.180

16. **Stone, M.,** Pope, A., McKinney, J., et al., 2022, *Measuring Star Formation and Black Hole Accretion Rates in Tandem using Mid-Infrared Spectra of Local Infrared-Luminous Galaxies.* ApJ 934.27

Major contributions:

15. Helton, J., ..., **Stone, M.**, et al. 2026, *Ionizing Photon Production Efficiencies and Chemical Abundance at Cosmic Dawn Revealed by Ultra-Deep Rest-Frame Optical Spectroscopy of JADES-GS-z14-0.*
ApJ 1004.30.

14. McKinney, J., ..., **Stone, M.** et al. 2026, *A JWST MIRI LRS Survey of 37 Massive Star-forming Galaxies and Active Galactic Nuclei at Cosmic Noon: Overview and First Results.* ApJ 1001.93

13. Sun, Y., Rieke, G., Lyu, J., **Stone, M.,** et al. 2025, *Evolution of the M_*–M_{BH} Relation from $z \sim 6$ to the Present Epoch.* ApJ 983.165

12. Lyu, J., Rieke, G., **Stone, M.,** et al., 2025, *Fading Light, Fierce Winds: JWST Snapshot of a Sub-Eddington Quasar at Cosmic Dawn.* ApJ 981.20

11. Goldberg, C., ..., **Stone, M.** et al. 2025, *Search for High-excitation Emission Lines from a Quasar-scale Active Nucleus in Arp 220.* ApJ 977.55

Co-authored:

10. De Rossi, M., ..., **Stone, M.** et al. 2026, *Determining Total Infrared Luminosities from Submillimeter Measurements of High-redshift Galaxies.* ApJ 1008.3

9. Helton, J., ..., **Stone, M.** et al. 2026, *The Stellar Populations and Rest-Frame Colors of Star-Forming Galaxies at $z \approx 8$: Exploring the Impact of Filter Choice and Star Formation History Assumption with JADES.* ApJ 1006.43

8. Eisenstein, D., ..., **Stone, M.** et al. 2026, *Overview of the JWST Advanced Deep Extragalactic Survey (JADES).* ApJS 283.6

7. Rieke, G., ..., **Stone, M.** et al. 2025, *Confirming Near- to Mid-infrared Photometrically Identified Obscured AGNs in the JWST Era.* ApJ 994.35

6. Rinaldi, P., ..., **Stone, M.** et al. 2025, *Not Just a Dot: The Complex UV Morphology and Underlying Properties of Little Red Dots.* ApJ 992.71

5. Rieke, G., ..., **Stone, M.** et al. 2025, *Low Accretion Rates in Black Holes in Late-stage Merger Ultraluminous Infrared Galaxies.* ApJ 988.17

4. Bonaventura, N., ..., **Stone, M.** et al. 2025, *The Relation between AGN and Host-galaxy Properties in the JWST Era. I. Seyferts at Cosmic Noon are Obscured and Disturbed.* ApJ 978.74

3. Sun, Y., ..., **Stone, M.** et al. 2025, *No Evidence for a Significant Evolution of M_*-M_{BH} Relation in Massive Galaxies up to $z \sim 4$.* ApJ 978.98

2. Alberts, S., ..., **Stone, M.** et al. 2024, *SMILES Initial Data Release: Unveiling the Obscured Universe with MIRI Multiband Imaging.* ApJ 976.224

1. Rieke, M., ..., **Stone, M.** et al. 2023, *JADES Initial Data Release for the Hubble Ultra Deep Field: Revealing the Faint Infrared Sky with Deep JWST NIRCam Imaging.* ApJS 269.16